The Traffic Signal Streak Analyzer

Week 1 Assignment - Problem 3 | Java

Problem Summary

Find the longest consecutive run of the same signal character and print its color and length. If longest streaks tie, the first one is retained.

Java Solution

```
public class Main {
    static void findLongestStreak(String signalLog) {
        if (signalLog == null || signalLog.isEmpty()) {
            System.out.println("No Signal Readings");
            return;
        }
        char bestColor = signalLog.charAt(0);
        int bestLength = 1;
        char currentColor = signalLog.charAt(0);
        int currentLength = 1;
        for (int i = 1; i < signalLog.length(); i++) {
            char c = signalLog.charAt(i);
            if (c == currentColor) currentLength++;
            else {
                currentColor = c;
                currentLength = 1;
            }
            if (currentLength > bestLength) {
                bestLength = currentLength;
                bestColor = currentColor;
            }
        }
        System.out.println("Longest Streak: '" + bestColor
            + "' repeated " + bestLength + " times");
    }
    public static void main(String[] args) {
        findLongestStreak("RRGGGYRR");
        findLongestStreak("RRRRYYGG");
    }
}
```

Sample Output

```
Longest Streak: 'G' repeated 3 times
Longest Streak: 'R' repeated 4 times
```

Complexity / Key Notes

Time: O(n) for a single pass. Extra space: O(1).